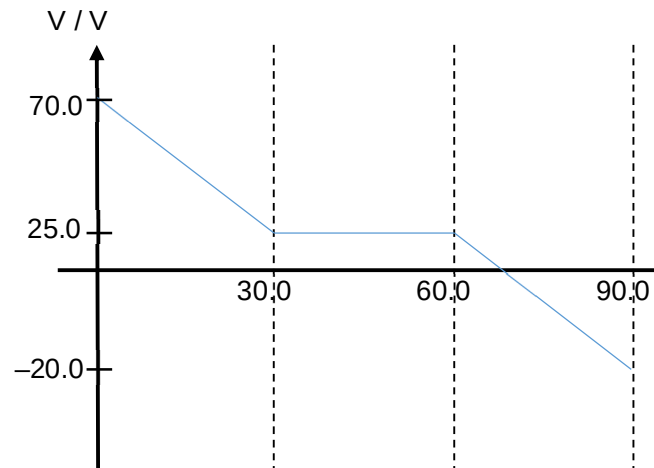


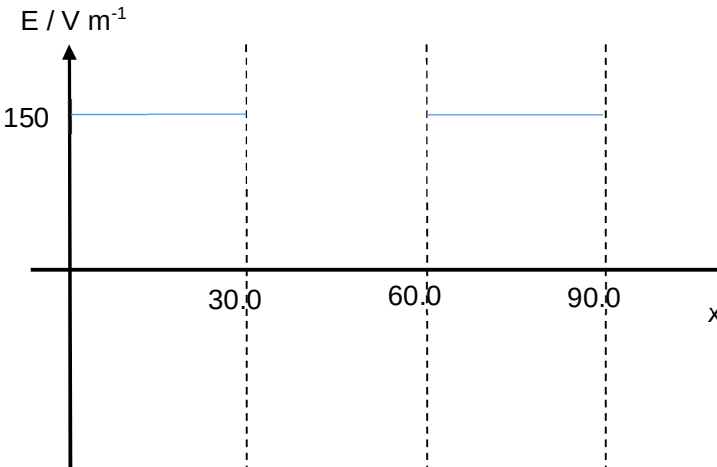
right coil is thus not magnetised.

8(a)

Electric field strength at a point is the electric force per unit positive test charge experienced by a charge placed at that point.

8(b)(i)



8(b)(ii)		
8(c)(i)	The electron would travel in a straight line or linear path in the direction of the resultant force towards plate A.	
8(c)(ii)	$a_x = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(100)}{9.11 \times 10^{-31}}$ $= 1.756 \times 10^{13} \text{ m s}^{-2}$ $S_x = u_x t + \frac{1}{2} a_x t^2$ $0.45 = \frac{1}{2} (1.756 \times 10^{13}) (t)^2$ $t = 2.26 \times 10^{-7} \text{ s}$ $v_x = u_x + a_x t$ $= (1.756 \times 10^{13}) (2.26 \times 10^{-7})$ $= 3.97 \times 10^6 \text{ m s}^{-1}$	